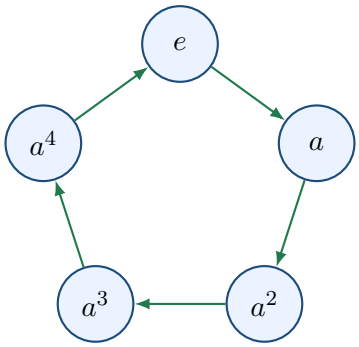
Lecture Notes on Group Theory

An introductory article-style handout



A cyclic group generated by one element

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1 Motivation and first examples

Group theory studies symmetry through algebra. A group is a set of objects together with a rule for composing them. The same formal language describes integers under addition, rotations of a polygon, invertible matrices, permutations of a finite set, and transformations of a geometric object.

The central idea is simple: instead of studying a single object directly, study the transformations that preserve its structure. This makes group theory a common language for algebra, geometry, number theory, physics, and coding theory.

Definition 1.1: group

A **group** is a pair $(G, *)$, where G is a set and $* : G \times G \to G$ is a binary operation, such that:

1. **Associativity:** $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$.
2. **Identity:** there exists $e \in G$ such that $e * a = a * e = a$ for all $a \in G$.
3. **Inverses:** for every $a \in G$ there exists $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

If, additionally, $a * b = b * a$ for all $a, b \in G$, then G is called **abelian**.

A compact way of reading the definition is: a group is a system in which one may compose operations, do nothing, and undo every operation.

Example 1.2: standard groups

- $(\mathbb{Z}, +)$ is an infinite abelian group. The identity is 0, and the inverse of n is $-n$.
- $(\mathbb{R} \setminus \{0\}, \cdot)$ is an infinite abelian group. The identity is 1, and the inverse of x is $1/x$.
- The set $M_n(\mathbb{R})$ of all real $n \times n$ matrices is not a group under multiplication, because not every matrix is invertible.
- The set $\mathrm{GL}_n(\mathbb{R})$ of invertible real $n \times n$ matrices is a group under multiplication. It is non-abelian for $n \geq 2$.
- The set S_n of all permutations of $\{1, \ldots, n\}$ is a group under composition.

1.1 How to prove that something is a group

A common proof pattern is to verify the axioms directly. In practice, the hardest parts are closure and inverses. Associativity often follows from a known operation, such as addition, multiplication, matrix multiplication, or function composition.

Common mistake

Closure is part of the meaning of a binary operation on G. If the operation can leave the set, then one does not have a group on that set. For example, positive integers are not a group under subtraction, because $2 - 5 = -3 \notin \mathbb{Z}_{>0}$.

2 Elementary consequences of the axioms

The group axioms look short, but they imply useful cancellation and uniqueness properties. These are often used silently in longer proofs.

Proposition 2.1: uniqueness of identity and inverses

Let G be a group.

1. The identity element is unique.
2. For each $a \in G$, the inverse a^{-1} is unique.
3. $(a^{-1})^{-1} = a$ and $(ab)^{-1} = b^{-1}a^{-1}$.

Proof. If e and f are both identities, then $e = ef = f$. If b and c are both inverses of a, then

$$b = be = b(ac) = (ba)c = ec = c.$$

The remaining formulas follow by checking that the proposed element satisfies the defining property of the inverse. For example,

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aa^{-1} = e.$$

□

Proposition 2.2: cancellation laws

For all $a, b, c \in G$,

$$ab = ac \Rightarrow b = c, \qquad ba = ca \Rightarrow b = c.$$

For instance, if $ab = ac$, multiply on the left by a^{-1}. This gives $a^{-1}ab = a^{-1}ac$, hence $b = c$.

2.1 Solving equations in groups

Every linear-looking group equation has a unique solution once the side of multiplication is handled correctly. The equation

$$ax = b$$

has solution $x = a^{-1}b$, while

$$xa = b$$

has solution $x = ba^{-1}$. In a non-abelian group these two expressions need not be equal.

Example 2.3: order matters

In matrix groups, AB usually differs from BA. For

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

one computes

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Thus $\mathrm{GL}_2(\mathbb{R})$ is not abelian.

3 Subgroups

A subgroup is a smaller group living inside a larger group. Subgroups are the first way to understand internal structure.

Definition 3.1: subgroup

Let G be a group. A subset $H \subseteq G$ is a **subgroup**, written $H \leq G$, if H is itself a group under the operation inherited from G.

The definition is conceptually clear but inefficient. The following criterion is the standard tool.

The one-step subgroup test

A nonempty subset $H \subseteq G$ is a subgroup if and only if

$$xy^{-1} \in H \qquad \text{for all } x, y \in H.$$

Proof idea. If H is a subgroup, the condition follows from closure and inverses. Conversely, choose $x \in H$. Then $xx^{-1} = e \in H$. Since $e, x \in H$, the condition gives $ex^{-1} = x^{-1} \in H$. Finally, if $x, y \in H$, then $y^{-1} \in H$, so $x(y^{-1})^{-1} = xy \in H$.

Example 3.2: subgroups of integers

Every subgroup of $(\mathbb{Z}, +)$ has the form $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ for a unique integer $n \geq 0$.

Reason. Let $H \leq \mathbb{Z}$. If $H = \{0\}$, then $H = 0\mathbb{Z}$. Otherwise, let n be the smallest positive element of H. Division with remainder gives $h = qn + r$ with $0 \leq r < n$. Since $h, qn \in H$, also $r = h - qn \in H$. Minimality forces $r = 0$, so $h \in n\mathbb{Z}$.

3.1 Generated subgroups

Given a subset $S \subseteq G$, the subgroup generated by S, denoted $\langle S \rangle$, is the smallest subgroup of G containing S. It consists of all finite products of elements of S and their inverses.

For one element $a \in G$,

$$\langle a \rangle = \{a^n : n \in \mathbb{Z}\}.$$

This set is always abelian even if G is not.

4 Cyclic groups and orders

Cyclic groups are the simplest groups: every element is a power of one generator.

Definition 4.1: cyclic group and order of an element

A group G is **cyclic** if $G = \langle a \rangle$ for some $a \in G$. The element a is called a **generator**.
The **order** of an element $a \in G$, denoted $\operatorname{ord}(a)$, is the smallest positive integer n such that $a^n = e$, if such an integer exists. If no such n exists, a has infinite order.

Proposition 4.2: structure of cyclic groups

If $G = \langle a \rangle$, then either $G \cong \mathbb{Z}$ or $G \cong \mathbb{Z}/n\mathbb{Z}$ for some $n \geq 1$. In particular, every cyclic group is abelian.

Example 4.3: residues modulo n

The additive group $\mathbb{Z}/n\mathbb{Z}$ is cyclic, generated by $\overline{1}$. An element $\overline{k}$ generates $\mathbb{Z}/n\mathbb{Z}$ if and only if $\gcd(k, n) = 1$.

Group	Operation	Identity	Typical generator
$\mathbb{Z}$	addition	0	1 or -1
$\mathbb{Z}/6\mathbb{Z}$	addition mod 6	$\overline{0}$	$\overline{1}, \overline{5}$
$\{1, -1, i, -i\}$	complex multiplication	1	i or $-i$
Rotations of a square	composition	no rotation	rotation by $90°$

4.1 A useful formula

If a has finite order n, then

$$a^k = e \quad \Longleftrightarrow \quad n \mid k.$$

Also, the order of a^k is

$$\operatorname{ord}(a^k) = \frac{n}{\gcd(n, k)}.$$

This formula is a quick way to count generators of a finite cyclic group.

5 Permutation groups

The symmetric group S_n consists of all bijections of the set $\{1, \ldots, n\}$ to itself. Its operation is composition. Since there are $n!$ bijections, $|S_n| = n!$.

Cycle notation

A cycle $(a_1\ a_2\ \ldots\ a_k)$ sends a_1 to a_2, a_2 to a_3, and so on, finally sending a_k back to a_1. Elements not listed are fixed.

For example, in S_5 the permutation

$$\sigma = (1\ 4\ 3)(2\ 5)$$

sends $1 \mapsto 4$, $4 \mapsto 3$, $3 \mapsto 1$, $2 \mapsto 5$, and $5 \mapsto 2$.

Theorem 5.1: disjoint cycle decomposition

Every permutation can be written as a product of disjoint cycles. This representation is unique up to the order of the cycles and cyclic rotation inside each cycle.

Disjoint cycles commute. Non-disjoint cycles usually do not commute.

Example 5.2: composition

Using right-to-left composition,

$$(1\ 2\ 3)(1\ 2) = (1\ 3), \qquad (1\ 2)(1\ 2\ 3) = (2\ 3).$$

The results differ, so S_3 is non-abelian.

5.1 Parity and the alternating group

Every permutation can be written as a product of transpositions. Although this expression is not unique, the parity of the number of transpositions is unique.

Definition 5.3: sign

The sign of a permutation σ is

$$\operatorname{sgn}(\sigma) = \begin{cases} 1, & \sigma \text{ is even,} \\ -1, & \sigma \text{ is odd.} \end{cases}$$

The alternating group is

$$A_n = \{\sigma \in S_n : \operatorname{sgn}(\sigma) = 1\}.$$

For $n \geq 2$, $|A_n| = n!/2$.

6 Dihedral groups

A second family of groups appears whenever we study symmetries of regular polygons. The dihedral group D_n is the group of all symmetries of a regular n-gon: n rotations and n reflections, so $|D_n| = 2n$.

Definition 6.1: presentation of D_n

The dihedral group of order $2n$ can be described by two generators:

$$D_n = \langle r, s : r^n = e,\ s^2 = e,\ srs = r^{-1} \rangle.$$

Here r is rotation by $2\pi/n$, and s is one reflection.

The relation $srs = r^{-1}$ says that reflection reverses orientation. It is the algebraic reason why D_n is usually non-abelian. For $n \geq 3$,

$$rs \neq sr.$$

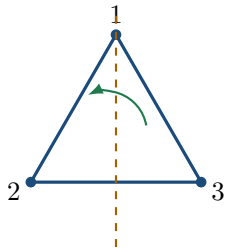
Indeed, from $srs = r^{-1}$ we get $sr = r^{-1}s$, so commuting would force $r = r^{-1}$.

Example 6.2: symmetries of a triangle

The group D_3 has six elements:

$$D_3 = \{e, r, r^2, s, rs, r^2s\}, \qquad r^3 = e, \quad s^2 = e.$$

It is isomorphic to S_3. The three rotations correspond to even permutations of the vertices, while the three reflections correspond to transpositions.



D_3: rotations and reflections of a triangle

Useful normal subgroup in D_n

The rotation subgroup

$$\langle r \rangle = \{e, r, r^2, \dots, r^{n-1}\}$$

is normal in D_n and has index 2. The quotient $D_n/\langle r \rangle$ is isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

This example is worth remembering: it shows how geometry, generators and relations, normal subgroups, and quotients meet in one compact object.

7 Cosets and Lagrange's theorem

Cosets are translates of a subgroup. They partition the whole group into equally sized pieces.

Definition 6.1: left and right cosets

Let $H \leq G$ and $a \in G$. The left coset of H by a is

$$aH = \{ah : h \in H\}.$$

The right coset is

$$Ha = \{ha : h \in H\}.$$

Theorem 6.2: Lagrange's theorem

If G is finite and $H \leq G$, then

$$|H| \mid |G|.$$

More precisely, $|G| = [G : H]|H|$, where $[G : H]$ is the number of left cosets of H in G.

Proof sketch. Each left coset aH has the same size as H, because the map $H \to aH$, $h \mapsto ah$, is a bijection. Distinct left cosets are disjoint, and all left cosets cover G. Therefore G is a disjoint union of $[G : H]$ subsets, each of size $|H|$.

Example 6.3: immediate consequences

Let G be finite.

- The order of every element $a \in G$ divides $|G|$, because $\operatorname{ord}(a) = |\langle a \rangle|$.
- If $|G| = p$ is prime, then G is cyclic and isomorphic to $\mathbb{Z}/p\mathbb{Z}$.
- There is no subgroup of order 4 in a group of order 10.

The converse is false

Lagrange's theorem says that subgroup orders divide the group order. It does not say that every divisor occurs as a subgroup order. For example, A_4 has order 12, but it has no subgroup of order 6.

8 Normal subgroups and quotient groups

Normal subgroups are exactly the subgroups that can be collapsed to form a quotient group.

Definition 7.1: normal subgroup

A subgroup $N \leq G$ is **normal**, written $N \trianglelefteq G$, if

$$gN = Ng \qquad \text{for every } g \in G.$$

Equivalently, $gng^{-1} \in N$ for every $g \in G$ and every $n \in N$.

Every subgroup of an abelian group is normal. In a non-abelian group, normality is an additional symmetry condition.

Example 7.2: normal and non-normal subgroups of S_3

In S_3, the subgroup $A_3 = \{e, (1\ 2\ 3), (1\ 3\ 2)\}$ is normal because it is the kernel of the sign map.
The subgroup $H = \{e, (1\ 2)\}$ is not normal. For instance,

$$(1\ 2\ 3)(1\ 2)(1\ 2\ 3)^{-1} = (2\ 3) \notin H.$$

8.1 Quotient groups

If $N \trianglelefteq G$, the quotient group G/N is the set of cosets gN with multiplication

$$(gN)(hN) = (gh)N.$$

Normality is exactly what makes this operation well-defined.

Theorem 7.3: quotient order

If G is finite and $N \trianglelefteq G$, then

$$|G/N| = \frac{|G|}{|N|}.$$

Quotients formalize the idea of identifying all elements of N with the identity. They are useful because they preserve part of the group structure while simplifying the original group.

9 Homomorphisms and isomorphisms

A homomorphism is a structure-preserving map between groups.

Definition 8.1: homomorphism

Let $(G, *)$ and $(H, \circ)$ be groups. A function $\varphi : G \to H$ is a **group homomorphism** if

$$\varphi(a * b) = \varphi(a) \circ \varphi(b)$$

for all $a, b \in G$.

Homomorphisms preserve identities, inverses, powers, and generated subgroups:

$$\varphi(e_G) = e_H, \qquad \varphi(a^{-1}) = \varphi(a)^{-1}, \qquad \varphi(a^n) = \varphi(a)^n.$$

Kernel and image

For a homomorphism $\varphi : G \to H$,

$$\mathrm{Ker}(\varphi) = \{g \in G : \varphi(g) = e_H\}, \qquad \mathrm{im}(\varphi) = \{\varphi(g) : g \in G\}.$$

Theorem 8.2: first isomorphism theorem

If $\varphi : G \to H$ is a group homomorphism, then $\mathrm{Ker}(\varphi) \trianglelefteq G$, $\mathrm{im}(\varphi) \leq H$, and

$$G/\mathrm{Ker}(\varphi) \cong \mathrm{im}(\varphi).$$

This theorem says that every homomorphism is, up to isomorphism, a quotient followed by an inclusion.

$$\begin{tikzcd} G \arrow[r, "\text{quotient}"] & G/\mathrm{Ker}\varphi \arrow[r, "\cong"] & \mathrm{im}\varphi \arrow[r, "\text{include}"] & H \end{tikzcd}$$

Example 8.3: determinant

The determinant map

$$\det : \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^\times$$

is a homomorphism because $\det(AB) = \det(A)\det(B)$. Its kernel is

$$\mathrm{SL}_n(\mathbb{R}) = \{A \in \mathrm{GL}_n(\mathbb{R}) : \det(A) = 1\}.$$

Thus $\mathrm{SL}_n(\mathbb{R})$ is a normal subgroup of $\mathrm{GL}_n(\mathbb{R})$.

10 Group actions

A group action is a formal way to say that a group acts by transformations on a set.

Definition 9.1: group action

A left action of a group G on a set X is a map

$$G \times X \to X, \qquad (g, x) \mapsto g \cdot x,$$

such that

$$e \cdot x = x, \qquad (gh) \cdot x = g \cdot (h \cdot x)$$

for all $g, h \in G$ and $x \in X$.

Actions connect abstract algebra with geometry and combinatorics. Examples include rotations acting on vertices of a polygon, matrices acting on vectors, and permutations acting on finite sets.

Orbit and stabilizer

For $x \in X$, the **orbit** of x is

$$Gx = \{g \cdot x : g \in G\}.$$

The **stabilizer** of x is

$$G_x = \{g \in G : g \cdot x = x\}.$$

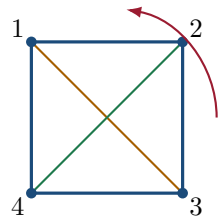
Theorem 9.2: orbit-stabilizer theorem

If G is finite and acts on X, then for every $x \in X$,

$$|G| = |Gx|\,|G_x|.$$

Example 9.3: rotations of a square

Let G be the rotation group of a square, so $|G| = 4$. Acting on the four vertices, the orbit of any vertex has size 4, and the stabilizer of a vertex has size 1. Acting on the two diagonals, the orbit of one diagonal has size 2, and its stabilizer has size 2.



rotation action on vertices and diagonals

11 Conjugacy and class equation

Conjugation measures how an element changes under a change of perspective inside the same group.

Definition 10.1: conjugacy

Elements $a, b \in G$ are **conjugate** if there exists $g \in G$ such that

$$b = gag^{-1}.$$

The conjugacy class of a is

$$\mathrm{Cl}(a) = \{gag^{-1} : g \in G\}.$$

The centralizer of a is

$$C_G(a) = \{g \in G : ga = ag\}.$$

Conjugacy classes are orbits for the action of G on itself by conjugation. Therefore orbit-stabilizer

gives

$$|\mathrm{Cl}(a)| = [G : C_G(a)].$$

The class equation

For a finite group G,

$$|G| = |Z(G)| + \sum [G : C_G(a_i)],$$

where $Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}$ is the center, and the sum runs over representatives of non-central conjugacy classes.

The class equation is a counting identity, but it has strong consequences. For example, every finite p-group has nontrivial center.

12 Direct products and abelian groups

Direct products build larger groups from smaller ones.

Definition 11.1: direct product

If G and H are groups, their direct product $G \times H$ is the set of pairs (g, h) with operation

$$(g_1, h_1)(g_2, h_2) = (g_1 g_2, h_1 h_2).$$

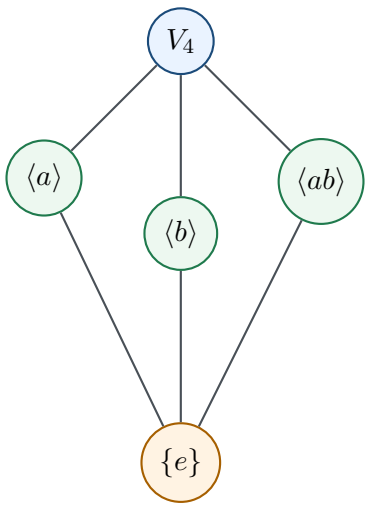
The identity is (e_G, e_H), and the inverse of (g, h) is (g^{-1}, h^{-1}). If both groups are finite, then $|G \times H| = |G||H|$.

Example 11.2: the Klein four group

The group

$$V_4 = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

has four elements, all non-identity elements have order 2, and it is not cyclic. It is isomorphic to the symmetry subgroup of a rectangle consisting of the identity, horizontal reflection, vertical reflection, and 180° rotation.



12.1 Finite abelian groups

A fundamental classification theorem says that every finite abelian group decomposes into a direct product of cyclic groups of prime-power order. A common version is

$$G \cong \mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}, \qquad n_1 \mid n_2 \mid \cdots \mid n_k.$$

This result is powerful because it reduces many questions about finite abelian groups to questions about cyclic groups.

13 Sylow theorems: a first orientation

The Sylow theorems are among the most important tools for finite groups. They describe how prime powers appear inside a finite group.

Let $|G| = p^a m$, where p is prime and $p \nmid m$. A subgroup of order p^a is called a Sylow p-subgroup.

Sylow theorem summary

Let G be finite and let p^a be the largest power of p dividing $|G|$.

1. G has a subgroup of order p^a.
2. Any two Sylow p-subgroups are conjugate.
3. If n_p is the number of Sylow p-subgroups, then

$$n_p \equiv 1 \pmod{p}, \qquad n_p \mid m.$$

Example 12.1: groups of order 15

Let $|G| = 15 = 3 \cdot 5$. The number n_5 of Sylow 5-subgroups satisfies

$$n_5 \equiv 1 \pmod{5}, \qquad n_5 \mid 3.$$

Thus $n_5 = 1$, so the Sylow 5-subgroup is normal. Similarly,

$$n_3 \equiv 1 \pmod{3}, \qquad n_3 \mid 5,$$

so the possibilities are 1 and 5; only 1 satisfies $n_3 \equiv 1 \pmod{3}$. Thus the Sylow 3-subgroup is also normal. The two normal cyclic subgroups have coprime orders, so $G \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \cong \mathbb{Z}/15\mathbb{Z}$.

14 A short problem set

The following questions are designed to test definitions, proof patterns, and computation.

Problem 1. Prove that the center $Z(G)$ is a subgroup of G.

Problem 2. Find all generators of the cyclic group $\mathbb{Z}/12\mathbb{Z}$.

Problem 3. Compute the order of $(1\ 2\ 3\ 4)(5\ 6)$ in S_6.

Problem 4. Let $\varphi : \mathbb{Z} \to \mathbb{Z}/8\mathbb{Z}$ be given by $\varphi(k) = \overline{3k}$. Find $\mathrm{Ker}(\varphi)$ and $\mathrm{im}(\varphi)$.

Problem 5. Prove that if $G/Z(G)$ is cyclic, then G is abelian.

Problem 6. Let G act on X. Prove that stabilizers are subgroups.

Problem 7. Show that every subgroup of a cyclic group is cyclic.

Selected solutions

Problem 2. The generators of $\mathbb{Z}/12\mathbb{Z}$ are exactly $\overline{k}$ with $\gcd(k, 12) = 1$, hence

$$\overline{1}, \overline{5}, \overline{7}, \overline{11}.$$

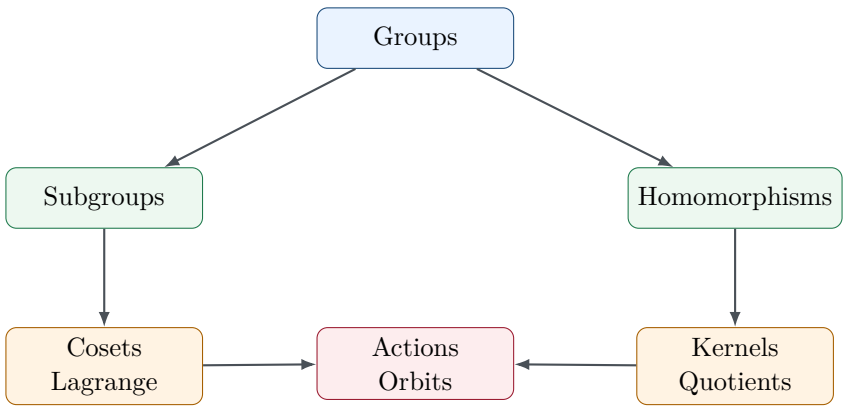
Problem 3. The disjoint cycles have lengths 4 and 2, so the order is

$$\operatorname{lcm}(4, 2) = 4.$$

Problem 4. Since 3 is invertible modulo 8, the image is all of $\mathbb{Z}/8\mathbb{Z}$. The kernel consists of all $k \in \mathbb{Z}$ such that $3k \equiv 0 \pmod 8$, equivalently $k \equiv 0 \pmod 8$. Hence $\operatorname{Ker}(\varphi) = 8\mathbb{Z}$.

Problem 5. Suppose $G/Z(G) = \langle gZ(G) \rangle$. Every element of G has the form $g^k z$ with $z \in Z(G)$. If $a = g^r z_1$ and $b = g^s z_2$, then the central factors commute with everything and the powers of g commute with each other. Hence $ab = ba$, so G is abelian.

15 Concept map and review checklist



Before moving to a second course in abstract algebra, make sure you can do the following without looking up definitions:

- verify group axioms in examples;
- use cancellation correctly;
- apply the subgroup test;
- compute orders in cyclic groups;
- work with cycle notation;
- compute cosets in small groups;
- apply Lagrange's theorem;
- test normality by conjugation;
- form basic quotient groups;
- find kernels and images;
- use the first isomorphism theorem;
- apply orbit-stabilizer;
- identify direct products;
- use Sylow congruences in small cases.

Final perspective

The most important habit is to distinguish three levels: elements, subsets, and maps. Elements have inverses and orders; subsets may be subgroups or normal subgroups; maps have kernels and images. Many proofs in group theory are short once the correct level is chosen.